Deepak Singh

+91 9978158483 | Jaipur, Rajasthan | deepaksingh91065@gmail.com | LinkedIn | GitHub

Summary

Final-year B.Tech student and SIH 2025 National Winner with hands-on experience building scalable full-stack and AI-powered applications using React, Node.js, Java, Spring Boot, Flutter, MongoDB, REST APIs, and Microservices. Passionate about software engineering, cloud technologies, and Generative AI.

Education

Modern Institute of Technology and Research Centre (MITRC), Alwar **(2022–2026)**
B.Tech in Computer Science, CGPA: 9.03

Kendriya Vidyalaya, Ahmedabad **(2021–2022)**
Class 12th (CBSE), 76%

Technical Skills

- **Languages:** JavaScript, TypeScript, Java, Python
- **Frontend:** React, Tailwind CSS, Flutter
- **Backend:** Node.js, Express.js, FastAPI, Spring Boot, JWT Auth, Docker
- **AI / ML & GenAI:** PyTorch, TensorFlow Lite, MobileNetV2, SGP4, LLM APIs, Prompt Engineering
- **Databases:** MySQL, MongoDB, PostgreSQL
- **Cloud & Tools:** Firebase, AWS, Azure, Vercel, Linux, Git, GitHub
- **Core Concepts:** DSA, OOP, DBMS, Operating Systems, REST APIs, CI/CD, Auth & Authorization

Projects

AgriMitra – Smart Farming & Crop Advisory Platform **GitHub | Live Site**
- Built a MERN-based PWA delivering multilingual crop advisory, weather insights, and market pricing.
- Integrated a FastAPI microservice running a PyTorch MobileNetV2 model for plant disease detection.
- Designed dashboards and REST APIs supporting real-time agricultural recommendations.

City Guardian – AI-Based Civic Monitoring App **GitHub | Live Site**
- Developed a Flutter + TensorFlow Lite application for real-time pothole detection with 90%+ accuracy.
- Integrated Firebase and Google Maps APIs to visualize and manage 100+ geo-tagged pothole reports.
- Implemented location-based reporting and image upload workflows for efficient civic issue tracking.

SAT-VEX – Satellite Tracking & Footprint Prediction System **GitHub | Live Site**
- Engineered a satellite tracking system using TLE + SGP4 with 3D visualization in CesiumJS.
- Integrated ML-based correction models trained on 500+ TLE samples, reducing prediction deviation by 10–15%.
- Developed interactive orbital footprint prediction and real-time satellite monitoring dashboards.

Experience

Full Stack Web Development Intern – CodeUnia **Remote | Jul 2025 – Aug 2025**
- Built AlgoQuest, a MERN-based platform for visualizing sorting, searching, and graph algorithms.
- Implemented 12+ interactive modules using React, Node.js, MongoDB, and REST APIs, improving algorithm visualization and learning experience.

Achievement

Winner – Smart India Hackathon (SIH) 2025 **National Level (50,000+ Teams)**
Live Project
- Built an AI-based CME risk-assessment and space-weather forecasting platform for ISRO's problem statement.
- Engineered a FastAPI-based ML backend for solar event classification and automated reporting workflows.

Certifications

Advanced Java Programming — Skill Oceans **Jun 2024**
Full Stack Web Development — Upflairs **May 2025**